

ISMAIL ABDUL KAREEM

AI Software Engineer & Full-Stack Developer

03303911285 | ismail233290@gmail.com | [buildwithismail.xyz](mailto:ismail233290@gmail.com) | github.com/IsmailAbdulkareem | Mehran Town, Karachi, Pakistan

PROFESSIONAL SUMMARY

AI Software Engineer and Full-Stack Developer with 2+ years building production-grade agentic AI systems, LLM-integrated web applications, and autonomous automation pipelines. Proficient in Python (FastAPI), Next.js, React, and TypeScript — with deep hands-on experience across the AI/LLM stack: OpenAI Agents SDK, Claude API, LangChain, Gemini, and Model Context Protocol (MCP). Experienced deploying containerised microservices to Kubernetes, Vercel, and Docker, with a track record of shipping real products for paying clients across real estate, legal, and food service verticals.

TECHNICAL SKILLS

Frontend Next.js, React, TypeScript, JavaScript, Tailwind CSS, HTML5, CSS3, Responsive Design

Backend Python, FastAPI, Node.js, REST API Design, Async APIs, Microservices

AI / Agentic OpenAI Agents SDK, Claude API, Gemini API, LangChain, MCP (Model Context Protocol), RAG Pipelines, LLM Integration, Prompt Engineering

Databases MongoDB, PostgreSQL, Sanity CMS, Vector Databases (Pinecone / pgvector)

DevOps / Cloud Docker, Kubernetes, Helm, Minikube, Vercel, GitHub Actions, CI/CD, AWS (basic)

Tools Git, GitHub, Spec-Driven Development, Kafka, Dapr, Hugging Face Spaces, Streamlit

EXPERIENCE

Freelance AI Software Engineer & Full-Stack Developer

Jan 2023 – Present

Governor Sindh Initiative for GenAI, Web3 & Metaverse | Karachi, Pakistan

- Delivered 5+ production web applications for SMB clients across construction, food services, and personal branding verticals using Next.js 14, TypeScript, and Tailwind CSS — reducing client content management overhead by integrating Sanity CMS with role-based editing workflows.
- Designed, built, and deployed LLM-powered chatbot websites for 3+ client verticals (real estate, restaurant, legal) on Vercel — integrating Claude API for natural language Q&A and automated lead capture flows that deliver visitor data to clients without manual follow-up.
- Developed AI customer support agents for SMB clients using Gemini API and OpenAI Agents SDK, handling FAQ resolution, intake forms, and task routing without human agents — reducing client support load.
- Built and shipped 2 agentic AI systems under hackathon deadlines: one featuring event-driven microservices (Kafka, Dapr), FastAPI, and Kubernetes/Minikube deployment with GitHub Actions CI/CD; another featuring multi-platform browser automation via Playwright and MCP tool integrations.
- Architected headless CMS integrations using Sanity and REST APIs, enabling non-technical clients to self-manage website content — eliminating recurring update requests and cutting post-launch support time by ~60%.
- Reduced page load times by up to 45% across client projects by implementing code splitting, lazy loading, and next/image optimisation in Next.js — directly improving Core Web Vitals scores and SEO rankings.

FEATURED PROJECTS

Personal Portfolio with AI Lead Capture buildwithismail.xyz

Stack: Next.js, Claude API, Vercel

- Production portfolio featuring a custom-built AI chatbot that answers questions about my work, collects visitor details through natural conversation, and automatically delivers captured leads — demonstrating real-world LLM integration beyond a static showcase.

Personal AI Employee [github.com/IsmailAbdulkareem/Personal AI Employee Gold Tier](https://github.com/IsmailAbdulkareem/Personal_AI_Employee_Gold_Tier)

Stack: Python, OpenAI Agents SDK, Playwright, MCP

- Fully autonomous social media agent using OpenAI Agents SDK, Playwright browser automation, and ModelContext Protocol (MCP) — capable of posting to LinkedIn and WhatsApp without human intervention, with persistent session management and a vault-based task queue architecture.
- Supports multi-platform watchers (LinkedIn, WhatsApp, Twitter, Instagram) with agentic decision-making and structured tool-use patterns.

AI Todo App todohackathonphase3.vercel.app

Stack: Next.js, OpenAI API, FastAPI, Kubernetes, Docker

- Shipped a production-grade AI task manager in a hackathon timeline — full microservices stack with FastAPI, Kafka, Dapr, Kubernetes (Minikube), and OpenAI API for NLP-driven task parsing and prioritisation; deployed via GitHub Actions CI/CD pipeline.

AI Chatbot Website Platform

Stack: Next.js, Claude API, Voiceflow, Make, Vercel, Stripe

- In-progress SaaS-style chatbot platform for SMBs across real estate, restaurants, and law firms — covering full product lifecycle: UI design, RAG pipeline, LLM integration, client onboarding, and Stripe payment processing.

EDUCATION

Governor Sindh Initiative for GenAI, Web3 & Metaverse — IT

Feb 2023 – Present

Google AI Professional Certificate (Coursera / TechValley Sindh)

In Progress 2026

Inter in Commerce — National College of Business Administration & Economics

Karachi, Pakistan

LANGUAGES

English (Professional) | Urdu (Native)